THE FOURIER ENTROPY–INFLUENCE INEQUALITY FOR SYMMETRIC BOOLEAN FUNCTIONS

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Abstract. We prove that every symmetric Boolean function f satisfies $\mathsf{H}[f] \le C_{\mathrm{sym}}\mathsf{I}[f]$ with $C_{\mathrm{sym}} = 2.692 + 2 + 1/\ln 2 < 6.1347$, improving the constant 12.04 of O'Donnell, Wright and Zhou. The Fourier entropy of a symmetric function splits into the entropy of its level weights and a within-level term. The first is at most $2.692\,\mathsf{I}[f]$ for every Boolean function, by a sharpened spectral-level estimate whose last step is a one-variable inequality checked by interval arithmetic. For the second we prove that the derivative of a symmetric function has noise stability at most $e^{-v}I_0(v)$ times its squared norm, where v is the variance of the noise, and evaluate the resulting entropy integral through a Frullani identity and two inequalities for modified Bessel functions. The AND and OR functions show that the optimal constant for symmetric functions is at least 4. We also prove that the entropy cost of restricting one coordinate of a symmetric function is asymptotically at most 4.637 times its influence, so the restriction step that would give constant 4.8723 can fail in only finitely many dimensions.

1. Introduction

For $f : \{-1,1\}^n \to \{-1,1\}$ write $f = \sum_{S\subseteq[n]} \widehat{f}(S)\chi_S$ with $\chi_S(x) = \prod_{i\in S} x_i$. By Parseval the squared coefficients form a probability distribution. Its entropy and the total influence are

$$\mathsf{H}[f] = \sum_S \widehat{f}(S)^2 \log_2 \frac{1}{\widehat{f}(S)^2}, \qquad \mathsf{I}[f] = \sum_S |S|\,\widehat{f}(S)^2 = \sum_{i=1}^n \mathbb{P}_x\big[f(x) \ne f(x^{\oplus i})\big],$$

where $x^{\oplus i}$ is x with its i-th coordinate flipped. The Fourier entropy–influence conjecture of Friedgut and Kalai [4] asserts that $\mathsf{H}[f] \le C\,\mathsf{I}[f]$ for an absolute constant C. It remains open. It is known for several classes of functions, among them symmetric functions [7, 5, 3, 1, 8].

A function is *symmetric* if $f(x)$ depends only on the Hamming weight $|x| = \#\{i : x_i = -1\}$. O'Donnell, Wright and Zhou [7] proved the conjecture for symmetric functions with constant 12.04. Our main result lowers this constant by about half.

Theorem A. *Every symmetric Boolean function f satisfies*

$$\mathsf{H}[f] \le C_{\mathrm{sym}}\,\mathsf{I}[f], \qquad C_{\mathrm{sym}} = \frac{673}{250} + 2 + \frac{1}{\ln 2} = 6.1346950408\ldots < 6.1347.$$

The proof has two parts. For symmetric f the coefficient $\widehat{f}(S)$ depends only on $k = |S|$, and the entropy splits as $\mathsf{H} = L + B$, where L is the entropy of the level weights $W_k = \sum_{|S|=k} \widehat{f}(S)^2$ and $B = \sum_k W_k \log_2 \binom{n}{k}$ measures the spread inside the levels. The bound $L \le \log_2(4+e)\mathsf{I} = 2.7481\mathsf{I}$ holds for every Boolean function; it is Lemma 2.1 of [8]. In Section 3 we lower it to $L \le 2.692\,\mathsf{I}$ by keeping the exact binary entropy and the exact isoperimetric bound in that argument; what remains is an inequality in one variable, which we verify by outward-rounded interval arithmetic. The main new part is the within-level bound $B < (2 + 1/\ln 2)\mathsf{I}$ for nonconstant f. It rests on the

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following noise-stability estimate, which is of independent interest. Write T_λ for the noise operator, $\mathrm{Stab}_\lambda[g] = \langle g, T_\lambda g\rangle = \sum_S \lambda^{|S|}\widehat{g}(S)^2$, and I_0 for the modified Bessel function of order zero.

Theorem B. *Let f be a symmetric Boolean function on $n \geq 2$ variables, $N = n-1$, and let $g = D_n f$ be its derivative in the last coordinate, a function on $\{-1,1\}^N$. For $0 \leq \lambda \leq 1$,*

$$\mathrm{Stab}_\lambda[g] \leq e^{-v} I_0(v)\, \|g\|_2^2, \qquad v = \frac{N(1-\lambda^2)}{4}.$$

Since $e^{-v}I_0(v) \sim (2\pi v)^{-1/2}$, this improves the derivative estimate used in [7] by a factor that tends to 2 when $\lambda \to 1$ and $v \to \infty$ together. The proof uses three facts: the derivative of a symmetric function is an alternating sum of indicators of Hamming slices; an alternating sum of selected values of a unimodal sequence is at most its largest value; and every atom of a sum of independent Bernoulli variables is bounded by Fourier inversion. The within-level term is then bounded by an integral of the envelope $e^{-v}I_0(v)$ against a harmonic-number kernel, which we show is less than $\ln 4$ for every N. A Frullani identity evaluates the main part exactly, and two inequalities for $e^{-x}I_0(x)$ and $e^{-x}I_1(x)$ control the rest.

The constant 4 is a natural lower bound.

Theorem C. *For the OR function on n variables (and for AND),*

$$\frac{\mathsf{H}[\mathrm{OR}_n]}{\mathsf{I}[\mathrm{OR}_n]} = 4 - \frac{4 - 2/\ln 2}{n} + O(2^{-n}).$$

Hence the optimal constant for symmetric functions lies in [4, 6.1347].

That OR approaches 4 is classical; we include the exact expression because AND and OR are the maximizers among all symmetric functions in every dimension we computed (Section 8).

The inductive approach to the conjecture asks for a coordinate whose restriction costs little entropy. For a coordinate i let $f_{i\to\pm1}$ be the two restrictions and

$$\Delta_i[f] = \mathsf{H}[f] - \tfrac{1}{2}\big(\mathsf{H}[f_{i\to1}] + \mathsf{H}[f_{i\to-1}]\big).$$

If $\Delta_1[f] \leq K\,\mathsf{I}_1[f]$ for every symmetric f, then $\mathsf{H} \leq K\mathsf{I}$ for every symmetric f by induction on n (Lemma 7.1). The best K is at least Δ_1/I_1 for OR_3, which equals $12 - \frac{9}{2}\log_2 3 = 4.8677\ldots$, so this route cannot reach 4; in all dimensions $n \leq 16$ it is the largest value. We prove that the restriction step is much cheaper in high dimension.

Theorem D.

$$\limsup_{n\to\infty}\ \sup_f\ \frac{\Delta_1[f]}{\mathsf{I}_1[f]} \leq C_* < 4.637,$$

where the supremum is over nonconstant symmetric f on n variables and

$$C_* = \frac{1}{\ln 2}\Big[\frac{14}{5} + \frac{89}{50}\,e^{-10}I_0(10) + \frac{48}{23\sqrt{40\pi}} + \frac{e^{-z}}{z}\Big] = 4.6364236659\ldots, \qquad z = 40\Big(\frac{23}{48}\Big)^2.$$

Consequently the restriction step with constant 4.8723 can fail for at most finitely many n. An explicit cutoff, together with a finite computation below it, would give $\mathsf{H} \leq 4.8723\,\mathsf{I}$ for every symmetric f. We do not have such a cutoff (Question 9.2).

The proof of Theorem D uses Theorem B in place of a log-Sobolev inequality, a passage to a limiting degree distribution, and one pointwise majorant of the restriction-entropy integrand that is verified by outward-rounded interval arithmetic. All other steps are analytic. The computations are in the repository listed under Data availability.

Notation. Logarithms are natural unless written $\log_2$. Entropies are in bits. For a symmetric function we write $F(w) \in \{-1,1\}$ for its value on weight w, $\widehat{f_k}$ for the common coefficient at level k, and $W_k = \binom{n}{k}\widehat{f_k}^2$. We put

$$\Phi(x) = e^{-x} I_0(x), \qquad G(x) = e^{-x} I_1(x) \qquad (x \geq 0),$$

so that $\Phi(x) = \frac{1}{\pi}\int_0^\pi e^{-x(1-\cos\theta)}\,d\theta$ and $\Phi' = G - \Phi$. Finally $\mathrm{E}_1(x) = \int_x^\infty e^{-u}u^{-1}\,du$ and $H_N = \sum_{i\leq N} 1/i$.

2. The entropy of a symmetric function

For symmetric f every set of size k carries the same coefficient, so

$$\mathsf{H}[f] = \sum_{k=0}^{n} \binom{n}{k}\widehat{f_k}^2 \log_2 \frac{1}{\widehat{f_k}^2} = \underbrace{\sum_k W_k \log_2 \frac{1}{W_k}}_{L} + \underbrace{\sum_k W_k \log_2 \binom{n}{k}}_{B}. \tag{1}$$

The level weights satisfy $\sum_k W_k = 1$ and $\sum_k kW_k = \mathsf{I}[f]$.

For $n \geq 2$ let $N = n-1$ and $g = D_n f$, where $D_n f(x) = \frac{1}{2}(f(x,1) - f(x,-1))$ for $x \in \{-1,1\}^N$. Then $\widehat{g}(T) = \widehat{f}(T \cup \{n\})$, so g is symmetric with

$$W_j[g] = \binom{N}{j}\widehat{f}^2_{j+1} = \frac{(j+1)W_{j+1}}{n}, \qquad \|g\|_2^2 = \sum_j W_j[g] = \frac{\mathsf{I}[f]}{n}. \tag{2}$$

Its values are $g(x) = \frac{1}{2}(F(w) - F(w+1))$ with $w = |x|$. They are nonzero exactly on the weights $w_0 < w_1 < \dots$ where F changes value, and there they alternate in sign. Up to a global sign,

$$g(x) = \sum_i (-1)^i \mathbf{1}[|x| = w_i]. \tag{3}$$

In particular $|g| = g^2$, so $\mathbb{E}|g| = \|g\|_2^2$.

3. The level entropy

Lemma 3.1 ([8, Lemma 2.1])**.** *Every Boolean function satisfies $L \leq \log_2(4+e)\,\mathsf{I}[f]$.*

Proof. We may assume f is not constant. Let $\varepsilon = 1 - W_0 > 0$ and $Q(k) = W_k/\varepsilon$ for $k \geq 1$. Put $a = \log_2(1 + e/4)$, so that $(2^a - 1)2^{-ak}$, $k \geq 1$, is a probability law. Gibbs' inequality against this law gives $\mathsf{H}(Q) \leq a\,\mathbb{E}_Q k - \log_2(2^a - 1)$. Since $L = h_2(\varepsilon) + \varepsilon \mathsf{H}(Q)$, $\varepsilon \mathbb{E}_Q k = \mathsf{I}$ and $h_2(\varepsilon) \leq \varepsilon\log_2(1/\varepsilon) + \varepsilon \log_2 e$, we get

$$L \leq a\mathsf{I} + \varepsilon\big[\log_2(1/\varepsilon) + \log_2 e - \log_2(e/4)\big] = a\mathsf{I} + \varepsilon\big[\log_2(1/\varepsilon) + 2\big].$$

Let $p = \min\{\mathbb{P}[f=1], \mathbb{P}[f=-1]\}$, so $\varepsilon = 4p(1-p)$. The edge-isoperimetric inequality [6, Chapter 2] gives $\mathsf{I} \geq 2p\log_2(1/p)$, and

$$2p\log_2\frac{1}{p} - \varepsilon\Big(1 + \tfrac{1}{2}\log_2\frac{1}{\varepsilon}\Big) = 2p\big[p\log_2 \tfrac{1}{p} + (1-p)\log_2(1-p)\big] \geq 0,$$

because the bracket is concave on $[0, \frac{1}{2}]$ (its second derivative is $(2p-1)/(p(1-p)\ln 2)$) and vanishes at both ends. Hence $\varepsilon[\log_2(1/\varepsilon) + 2] \leq 2\mathsf{I}$ and $L \leq (a+2)\mathsf{I} = \log_2(4+e)\mathsf{I}$. □

The proof loses in two places: it replaces $h_2(\varepsilon)$ by $\varepsilon\log_2(e/\varepsilon)$, and it bounds $\varepsilon[\log_2(1/\varepsilon)+2]$ by $2\mathsf{I}$ instead of dividing by the isoperimetric bound $2p\log_2(1/p)$ itself. Keeping both steps exact gives a smaller constant. Write $h(x) = -x\ln x - (1-x)\ln(1-x)$.

Lemma 3.2. *Every Boolean function satisfies $L \leq \frac{673}{250}\,\mathsf{I}[f] = 2.692\,\mathsf{I}[f]$.*

Proof. Assume f is not constant and let p, $\varepsilon = 4p(1-p)$ and Q be as in the previous proof, so $0 < p \le \frac{1}{2}$. Put $r = \frac{147}{100}$. Gibbs' inequality against the law $(r-1)r^{-k}$, $k \ge 1$, gives $-\sum_k Q(k)\ln Q(k) \le -\ln(r-1) + \mathbb{E}_Q k \,\ln r$, hence

$$L \le \log_2 r \cdot \mathsf{I} + \frac{h(\varepsilon) - \varepsilon \ln(r-1)}{\ln 2}.$$

The numerator is positive because $r < 2$. With $\mathsf{I} \ge 2p\log_2(1/p)$ this gives $L \le F(p)\,\mathsf{I}$, where

$$F(p) = \log_2 r + \frac{h(4p(1-p)) - 4p(1-p)\ln(r-1)}{2p\ln(1/p)}.$$

It remains to show $F(p) < 2.692$ on $0 < p \le \frac{1}{2}$. Write $q = \log_2(1/p) \ge 1$.

If $1 \le q \le 2$, that is $\frac{1}{4} \le p \le \frac{1}{2}$, then $2p\log_2(1/p) \ge 1$ by concavity, since it equals 1 at both ends. With $h \le \ln 2$ and $\varepsilon \le 1$ we get $L \le \big(\log_2 r + 1 - \log_2(r-1)\big)\mathsf{I} < 2.646\,\mathsf{I}$.

If $q \ge 32$, use $h(x) \le x\big(\ln(1/x)+1\big)$ and $\ln(1/\varepsilon) = q\ln 2 - \ln 4 - \ln(1-p)$:

$$F(p) \le \log_2 r + 2(1-p)\Big[1 + \frac{c(p)}{q\ln 2}\Big], \qquad c(p) = 1 - \ln 4 - \ln(1-p) - \ln(r-1).$$

Here $0 < c(p) \le c(2^{-32})$, so $F(p) \le \log_2 r + 2 + 2c(2^{-32})/(32\ln 2) < 2.590$.

For $2 \le q \le 32$ we evaluate F in outward-rounded interval arithmetic (45 digits) on a dyadic partition of $[2,32]$ in the variable q. The partition has 7058 intervals, they cover $[2,32]$ without gaps, and the upper end of every enclosure is below 2.692; the largest is 2.6919998948. □

The choice of r is close to optimal for this argument: on a fine grid $\sup_p F(p)$ is 2.6917 for $r = 1.47$ and for $r = 1.48$, and larger for $r = 1.44$ and $r = 1.50$. So the method stops near 2.6917.

4. Noise stability of derivatives

Lemma 4.1. *Let $(a_y)_{y\ge 0}$ be a nonnegative unimodal sequence and $y_0 < y_1 < \cdots < y_m$ integers. Then $\big|\sum_i (-1)^i a_{y_i}\big| \le \max_y a_y$.*

Proof. Write $a_y = \int_0^\infty \mathbf{1}[a_y > t]\,dt$. For each t the set $\{y : a_y > t\}$ is an interval, so the selected indices in it form a block of consecutive i, whose alternating sum is 0 or ± 1. The integrand vanishes for $t \ge \max a$. Integrate. □

Lemma 4.2. *Let Z be a sum of N independent Bernoulli variables with parameters in $\{p, 1-p\}$, and $v = Np(1-p)$. Then Z has a unimodal law and $\mathbb{P}[Z = y] \le \Phi(v)$ for every integer y.*

Proof. Each Bernoulli law is log-concave and convolution preserves log-concavity of sequences supported on intervals, so the law of Z is log-concave and in particular unimodal. Each summand has characteristic function of modulus $\big(1 - 4p(1-p)\sin^2(\theta/2)\big)^{1/2}$. Using $1 - u \le e^{-u}$,

$$|\mathbb{E}e^{i\theta Z}| \le \exp\big(-2v\sin^2(\theta/2)\big) = e^{-v(1-\cos\theta)}.$$

Fourier inversion gives $\mathbb{P}[Z = y] \le \frac{1}{2\pi}\int_{-\pi}^{\pi} |\mathbb{E}e^{i\theta Z}|\,d\theta \le \Phi(v)$. □

Proof of Theorem B. Let $p = (1-\lambda)/2$. Given x with $|x| = r$, the weight of a λ-correlated copy y of x is $Z_r = \mathrm{Bin}(r, 1-p) + \mathrm{Bin}(N-r, p)$, of variance $Np(1-p) = v$. By (3),

$$\mathrm{Stab}_\lambda[g] = \mathbb{E}_x\Big[g(x)\sum_i (-1)^i\,\mathbb{P}\big[Z_{|x|} = w_i\big]\Big].$$

By Lemmas 4.1 and 4.2 the inner sum has absolute value at most $\Phi(v)$. Hence $\mathrm{Stab}_\lambda[g] \le \Phi(v)\,\mathbb{E}|g| = \Phi(v)\|g\|_2^2$. □

Remark 4.3. For even N the atom bound before the step $1 - u \le e^{-u}$ is attained at $r = y = N/2$. The constant $\sup_v \sqrt{v}\,\Phi(v) = 0.46882\ldots$ is the sharp uniform concentration constant for Bernoulli sums of Baillon, Cominetti and Vaisman [2]; we use the full function Φ rather than this constant.

5. The within-level entropy

Lemma 5.1. *Let f be symmetric on $n \ge 2$ variables with $\mathsf{I}[f] > 0$, and $N = n-1$. Then*

$$B \le \frac{1 + C_N}{\ln 2}\,\mathsf{I}[f], \qquad C_N = \int_0^1 \frac{\Phi\big(Nt(2-t)/4\big) - (1-t)^N}{t}\,dt.$$

Proof. Using $\binom{n}{k} \le (en/k)^k$,

$$B \ln 2 \le \sum_{k\ge 1} k W_k\Big(1 + \ln\frac{n}{k}\Big) = \mathsf{I}\Big(1 + \mathbb{E}_q \ln\frac{n}{j+1}\Big), \qquad q_j = \frac{(j+1)W_{j+1}}{\mathsf{I}},\ 0 \le j \le N.$$

By (2), $q_j = W_j[g]/\|g\|_2^2$. Comparing the sum with an integral, $\ln\frac{n}{j+1} \le H_N - H_j = \int_0^1 \frac{\rho^j - \rho^N}{1-\rho}\,d\rho$. Therefore

$$\mathbb{E}_q \ln\frac{n}{j+1} \le \int_0^1 \frac{s(\rho) - \rho^N}{1-\rho}\,d\rho, \qquad s(\rho) = \sum_j q_j \rho^j = \frac{\mathrm{Stab}_\rho[g]}{\|g\|_2^2}.$$

By Theorem B, $s(\rho) \le \Phi\big(N(1-\rho^2)/4\big)$. Substituting $\rho = 1-t$, so that $1-\rho^2 = t(2-t)$, gives the claim. The integrand of C_N is $O(1)$ near $t = 0$. □

We need three facts about Φ and G.

Lemma 5.2. (i) *$\sqrt{w}\,\Phi(w) > 1/\sqrt{2\pi}$ for every $w \ge \frac12$.*

(ii) *$J(X) := \int_0^X u\,(-\Phi'(u))\,du = XG(X)$ for every $X \ge 0$.*

(iii) *$G(X) < 1/\sqrt{2\pi X}$ for every $X > 0$.*

Proof. (i) Let $h(w) = \sqrt{w}\,\Phi(w)$. Substituting $I_0 = e^w\Phi$ into the Bessel equation $I_0'' + I_0'/w - I_0 = 0$ gives

$$h'' + 2h' + \frac{h}{4w^2} = 0.$$

At a stationary point $h'' = -h/(4w^2) < 0$, so h has no local minimum in $(0,\infty)$. At the left end $h(\frac12) \ge e^{-1/2}/\sqrt{2} > 1/\sqrt{2\pi}$, since $I_0 \ge 1$ and $\pi > e$. As $w \to \infty$, the substitution $\theta = y/\sqrt{w}$ and dominated convergence (with $1 - \cos\theta \ge 2\theta^2/\pi^2$ on $[0,\pi]$) give $h(w) \to \frac{1}{\pi}\int_0^\infty e^{-y^2/2}dy = 1/\sqrt{2\pi}$. If $h(w_0) \le 1/\sqrt{2\pi}$ for some $w_0 > \frac12$, then h restricted to some $[\frac12, W]$ with $W > w_0$ attains its minimum at an interior point: if $h(w_0) < 1/\sqrt{2\pi}$ choose W with $h(W) > h(w_0)$, and if equality holds take any $W > w_0$, since then $h \ge h(w_0)$ on $[\frac12,\infty)$ by the first case. This interior minimum is a local minimum, a contradiction.

(ii) From $I_0' = I_1$ and $I_1' = I_0 - I_1/x$ we get $\Phi' = G - \Phi$ and $G' = \Phi - G - G/x$. Hence $(xG)' = x(\Phi - G) = -x\Phi'$, and $xG(x) \to 0$ as $x \to 0$.

(iii) Integrating $I_1(X) = \frac{1}{\pi}\int_0^\pi e^{X\cos\theta}\cos\theta\,d\theta$ by parts and substituting $u = 1 - \cos\theta$,

$$G(X) = \frac{X}{\pi}\int_0^\pi e^{-X(1-\cos\theta)}\sin^2\theta\,d\theta = \frac{X}{\pi}\int_0^2 e^{-Xu}\sqrt{u(2-u)}\,du$$
$$< \frac{\sqrt{2}X}{\pi}\int_0^\infty e^{-Xu}\sqrt{u}\,du = \frac{1}{\sqrt{2\pi X}}.$$ □

Proposition 5.3. *$C_N < \ln 4$ for every integer $N \ge 1$. More precisely $C_1 \le 1$, $C_2 \le 59/64$, and for $N \ge 3$*

$$C_N < \ln 4 - \frac{2-\sqrt{2}}{\sqrt{\pi N}} + \frac{1}{2N} < \ln 4.$$

Proof. Split $C_N = T_1 + T_2 + T_3$ with

$$T_1 = \int_0^1 \frac{\Phi(\frac{Nt(2-t)}{4}) - \Phi(\frac{Nt}{2})}{t}dt, \quad T_2 = \int_0^1 \frac{\Phi(\frac{Nt}{2}) - e^{-Nt}}{t}dt, \quad T_3 = \int_0^1 \frac{e^{-Nt} - (1-t)^N}{t}dt.$$

The term T_2. Write $\Phi(x) = \mathbb{E}_\theta e^{-x(1-\cos\theta)}$ with θ uniform on $[0,\pi]$. For each θ, Frullani's integral gives $\int_0^\infty \big(e^{-Nt(1-\cos\theta)/2} - e^{-Nt}\big)\frac{dt}{t} = \ln\frac{2}{1-\cos\theta}$; the integrand is nonnegative, so by Tonelli

$$\int_0^\infty \frac{\Phi(Nt/2) - e^{-Nt}}{t}dt = \mathbb{E}_\theta \ln\frac{1}{\sin^2(\theta/2)} = -\frac{4}{\pi}\int_0^{\pi/2} \ln\sin u\, du = \ln 4.$$

On $t \ge 1$ we have $Nt/2 \ge \frac{1}{2}$, so Lemma 5.2(i) gives $\int_1^\infty \Phi(Nt/2)\frac{dt}{t} \ge \int_1^\infty \frac{dt}{\sqrt{\pi N}\, t^{3/2}} = \frac{2}{\sqrt{\pi N}}$. Since $\int_1^\infty e^{-Nt}\frac{dt}{t} = \mathrm{E}_1(N)$,

$$T_2 \le \ln 4 - \frac{2}{\sqrt{\pi N}} + \mathrm{E}_1(N).$$

The term T_1. The function Φ is completely monotone, so $-\Phi'$ is positive and decreasing. With $a = Nt(2-t)/4$ and $b = Nt/2$ we have $a \ge Nt/4$ and $b - a = Nt^2/4$, so $\Phi(a) - \Phi(b) \le \frac{Nt^2}{4}\big(-\Phi'(\frac{Nt}{4})\big)$. By Lemma 5.2(ii) and (iii),

$$T_1 \le \int_0^1 \frac{Nt}{4}\Big(-\Phi'\big(\tfrac{Nt}{4}\big)\Big)dt = \frac{4}{N}J\big(\tfrac{N}{4}\big) = G\big(\tfrac{N}{4}\big) < \sqrt{\frac{2}{\pi N}}.$$

The term T_3. Since $\int_0^1 \frac{1-(1-t)^N}{t}dt = H_N$ and $\int_0^1 \frac{1-e^{-Nt}}{t}dt = \ln N + \gamma + \mathrm{E}_1(N)$,

$$T_3 = H_N - \ln N - \gamma - \mathrm{E}_1(N) \quad (\ge 0).$$

Adding, the $\mathrm{E}_1(N)$ terms cancel and $H_N - \ln N - \gamma < 1/(2N)$ gives

$$C_N < \ln 4 + \frac{\sqrt{2}}{\sqrt{\pi N}} - \frac{2}{\sqrt{\pi N}} + \frac{1}{2N}.$$

The last two terms together are negative when $N > \pi/(4(2-\sqrt{2})^2)$, and this threshold is less than 3 because $\pi < 4 < 12(2-\sqrt{2})^2$, the last inequality following from $\sqrt{2} < 17/12$.

For $N = 1$, $\Phi \le 1$ gives $C_1 \le \int_0^1 \frac{1-(1-t)}{t}dt = 1$. For $N = 2$, $e^{-z} \le 1 - z + z^2/2$ together with $\mathbb{E}(1-\cos\theta) = 1$ and $\mathbb{E}(1-\cos\theta)^2 = \frac{3}{2}$ gives $\Phi(v) \le 1 - v + \frac{3}{4}v^2$. With $v = (1-\rho^2)/2$,

$$C_2 \le \int_0^1 \frac{\frac{1-\rho^2}{2} + \frac{3}{16}(1-\rho^2)^2}{1-\rho}\, d\rho = \frac{3}{4} + \frac{3}{16}\cdot\frac{11}{12} = \frac{59}{64}.$$

Both are below $\ln 4$. □

Proof of Theorem A. For $n = 1$ every Boolean function is constant or $\pm x_1$ and $\mathsf{H} = 0$. For $n \ge 2$ and $\mathsf{I} > 0$, Lemma 5.1 and Proposition 5.3 give $B < \frac{1+\ln 4}{\ln 2}\mathsf{I} = (2 + \frac{1}{\ln 2})\mathsf{I}$, and Lemma 3.2 gives $L \le \frac{673}{250}\mathsf{I}$. Add them using (1). □

Remark 5.4. The two parts are not tight together. For majority on $n = 2m+1$ variables, $B/\mathsf{I} \to 2/\ln 2$, so the within-level constant cannot go below 2.885; and $L/\mathsf{I} \le 2.52$ for every symmetric function with $n \le 16$ (Section 8), below the 2.692 of Lemma 3.2. Numerically C_N increases to $\ln 4$: $C_1 = 0.6638$, $C_2 = 0.8911$, $C_3 = 0.9991$, $C_{10^6} = 1.3855$.

The majority limit in the remark is proved as follows. Write $n = 2m+1$. The derivative of majority is the indicator of the middle slice of the $2m$-cube, and its normalized spectral weights are

$$q_{2j} = \binom{2j}{j}\binom{2m-2j}{m-j}4^{-m}, \qquad q_{2j+1} = 0.$$

Thus j has the beta-binomial law with parameters $(m, \frac{1}{2}, \frac{1}{2})$, and j/m converges in law to the arcsine law on $(0,1)$. By (2), B/I is the expectation of $\log_2\binom{2m+1}{2j+1}/(2j+1)$. Stirling's formula gives the interior limit $h(x)/(x\ln 2)$ with $h(x) = -x\ln x - (1-x)\ln(1-x)$. The estimates $q_{2j} \le C/\sqrt{(j+1)(m-j+1)}$ and $\ln\binom{n}{k}/k \le 1 + \ln(n/k)$ bound the contribution of $j \le \delta m$ by

$C\sqrt{\delta}(1+\ln(1/\delta))+O(\ln(m)/\sqrt{m})$, and the other end contributes less, so the expectations converge. With $x=\sin^2 u$,

$$\mathbb{E}\frac{h(X)}{X}=\frac{2}{\pi}\int_0^{\pi/2}\left[-\ln(\sin^2 u)-\cot^2 u\ln(\cos^2 u)\right]du=2,$$

by $\int_0^{\pi/2}\ln\sin u\,du=-\frac{\pi}{2}\ln 2$ and $\int_0^{\pi/2}\cot^2 u\ln(\cos^2 u)\,du=\pi(\ln 2-1)$. Hence $B/\mathsf{I}\to 2/\ln 2$.

6. AND and OR

Proof of Theorem C. Let f take the value -1 at exactly one point; this is OR or AND up to the signs of the inputs and the output, which change neither H nor I. Then $\widehat{f}(\emptyset)=1-2^{1-n}$ and $\widehat{f}(S)^2=4^{1-n}$ for $S\neq\emptyset$. Hence, with $W_0=(1-2^{1-n})^2$,

$$\mathsf{H}[f]=W_0\log_2\frac{1}{W_0}+(2^n-1)\,4^{1-n}\cdot 2(n-1),\qquad \mathsf{I}[f]=n\,2^{1-n}.$$

The second term divided by I is $(2-2^{1-n})\frac{2(n-1)}{n}=4-\frac{4}{n}+O(2^{-n})$, and $W_0\log_2(1/W_0)=\frac{2^{2-n}}{\ln 2}(1+O(2^{-n}))$ contributes $\frac{2}{n\ln 2}+O(2^{-n})$. □

At $n=1024$ the ratio is $3.998911513751\ldots$.

7. The restriction step

Lemma 7.1. *Suppose $\Delta_1[f]\le K\,\mathsf{I}_1[f]$ for every symmetric f on at most m variables. Then $\mathsf{H}[f]\le K\,\mathsf{I}[f]$ for every symmetric f on at most m variables.*

Proof. Restrictions of a symmetric function are symmetric. For $i\neq 1$ the influence of i is the average of its influences in the two restrictions of x_1, so $\mathsf{I}[f]=\mathsf{I}_1[f]+\frac{1}{2}(\mathsf{I}[f_{1\to 1}]+\mathsf{I}[f_{1\to -1}])$. Induct on n, starting from constants. □

For symmetric f on n variables, the restrictions to $x_1=\pm 1$ have coefficients $\widehat{f}_s\pm\widehat{f}_{s+1}$ on the $\binom{N}{s}$ sets of size s in the remaining coordinates, $N=n-1$. With $\varphi(u)=u\log_2(1/u)$ this gives the exact formulas

$$\Delta_1=\sum_{s=0}^{N}\binom{N}{s}\mathcal{D}(\widehat{f}_s,\widehat{f}_{s+1}),\qquad \mathsf{I}_1=\sum_{s=0}^{N}\binom{N}{s}\widehat{f}_{s+1}^2,\tag{4}$$

$$\mathcal{D}(a,b)=\varphi(a^2)+\varphi(b^2)-\tfrac{1}{2}\left[\varphi((a+b)^2)+\varphi((a-b)^2)\right].$$

For OR_3 one finds $\Delta_1/\mathsf{I}_1=12-\frac{9}{2}\log_2 3$.

Proof of Theorem D. We work in natural units, $\mathcal{D}_e=(\ln 2)\mathcal{D}$, and divide by $\ln 2$ at the end.

Step 1: a tangent bound. Let $h(t)=\mathcal{D}_e(\sqrt{t},1)$ for $t\ge 0$. For $t>0$, $t\neq 1$, direct differentiation gives

$$h''(t)=\frac{1}{2t^{3/2}}\ln\Big|\frac{\sqrt{t}-1}{\sqrt{t}+1}\Big|<0,$$

h' is continuous at $t=1$ and tends to 0 from above, so h is increasing and concave and $(x,y)\mapsto\mathcal{D}_e(\sqrt{x},\sqrt{y})=y\,h(x/y)$ is concave on the quadrant. Every tangent gives $\mathcal{D}_e(a,b)\le h'(t)a^2+[h(t)-th'(t)]b^2$. Let $\mu_s=\binom{N}{s}\widehat{f}_{s+1}^2/\mathsf{I}_1$, a probability law on $\{0,\ldots,N\}$. Taking the tangent at $t_s=(n-s)/s$ for the term $s\ge 1$ of (4), and using $\binom{N}{s+1}/\binom{N}{s}=(N-s)/(s+1)$ to move the a^2 part to the neighbouring index, one obtains

$$\frac{(\ln 2)\Delta_1}{\mathsf{I}_1}\le\frac{\mathcal{D}_e(\widehat{f}_0,\widehat{f}_1)}{\mathsf{I}_1}+\mu_0\kappa(\tfrac{1}{n})+\sum_{s=1}^{N}\mu_s\Big[\Psi(\tfrac{s}{n})+\kappa(\tfrac{s+1}{n})-\kappa(\tfrac{s}{n})\Big],$$

where $\Psi(x) = h(\frac{1-x}{x})$ and $\kappa(x) = \frac{1-x}{x}h'(\frac{1-x}{x})$, extended continuously by $\kappa(0) = 1$, $\kappa(1) = 0$. Since κ is uniformly continuous on $[0,1]$, the sum of the κ-differences tends to 0 uniformly in f as $n \to \infty$.

We also use $h(t) \le 3 + \ln(1+t)$. With $p(y) = y^2 \ln(y^2)$, $p(0) = 0$, the second-difference identity (valid since p'' is locally integrable) and Jensen's inequality for the triangular density $1 - |u|$ on $[-1,1]$, which has mean 0 and second moment $\frac{1}{6}$, give

$$h(t) = 3 + \int_{-1}^{1} (1-|u|) \ln\big((\sqrt{t}+u)^2\big)\, du \le 3 + \ln\big(t + \tfrac{1}{6}\big) \le 3 + \ln(1+t).$$

Step 2: three facts about $g = D_1 f$. (a) Let $m = |\widehat{f_1}| = |\mathbb{E}g|$. Since $|g| = g^2$, $m \le \mathsf{I}_1$. By (3) and Lemma 4.1 applied to the binomial weights, $m \le b_N := \max_j \binom{N}{j} 2^{-N} = O(N^{-1/2})$. From $h(t) \le 3 + \ln(1+t)$ and $|\widehat{f_0}| \le 1$ one gets $\mathcal{D}_e(\widehat{f_0}, \widehat{f_1})/\mathsf{I}_1 \le b_N[3 + \ln(1 + b_N^{-2})] \to 0$, and $\mu_0 = m^2/\mathsf{I}_1 \le b_N \to 0$; when $m = 0$ these terms vanish. (b) Consecutive values of g on adjacent weights have nonpositive product, so $\|D_j g\|_2^2 \ge \frac{1}{2}\|g\|_2^2$ for each j, and summing gives $\mathbb{E}_\mu s \ge N/2$. (c) By Theorem B, $\sum_s \mu_s \lambda^s \le \Phi\big(N(1-\lambda^2)/4\big)$ for $0 \le \lambda \le 1$.

Step 3: the limit law. Take any sequence $n \to \infty$ with nonconstant symmetric f_n, and pass to a subsequence along which the law of s/n under μ converges weakly to a law ν on $[0,1]$. By (b), $\mathbb{E}_\nu X \ge \frac{1}{2}$. Taking $\lambda = e^{-t/n}$ in (c) and letting $n \to \infty$,

$$\mathbb{E}_\nu e^{-tX} \le \Phi(t/2) \qquad (t > 0), \tag{5}$$

and in particular $\nu(\{0\}) = 0$. The function Ψ is unbounded at 0 but satisfies $\Psi(x) \le 3 + \ln(1/x)$, and (c) with $\lambda = e^{-1/m}$ gives $\mu[s \le m] \le e\,\Phi\big(N(1-e^{-2/m})/4\big) \le C\sqrt{m/N}$ for an absolute C. Summing over dyadic ranges, the contribution of $1 \le s \le \delta n$ is $O(\sqrt{\delta}(1 + \ln(1/\delta)))$ uniformly in n. This uniform integrability, Step 2(a) and weak convergence give $\sum_{s \ge 1} \mu_s \Psi(s/n) \to \int \Psi\, d\nu$. Hence $\limsup(\ln 2)\Delta_1/\mathsf{I}_1 \le \sup_\nu \int \Psi\, d\nu$, the supremum over laws on $(0,1]$ with $\mathbb{E}_\nu X \ge \frac{1}{2}$ and (5).

Step 4: a Laplace majorant. For $0 < x \le 1$,

$$\Psi(x) \le U(x) := \mathrm{E}_1(40x) + \frac{553}{100} - \frac{273}{50}x + \frac{89}{50}e^{-20x}. \tag{6}$$

This is verified by outward-rounded interval arithmetic on 11,900 cells covering $[10^{-4}, 1]$, using that Ψ and U are decreasing (Appendix A), with minimum certified slack above 0.02288; for $0 < x \le 10^{-4}$ one uses $\Psi(x) \le 3 - \ln x$, $\mathrm{E}_1(40x) \ge -\gamma - \ln(40x)$ and $e^{-20x} \ge 1 - 20x$. Since $\mathrm{E}_1(40x) = \int_{40}^{\infty} e^{-tx}\frac{dt}{t}$, Tonelli, (5) and $\mathbb{E}_\nu X \ge \frac{1}{2}$ give

$$\int \Psi\, d\nu \le \int_{40}^{\infty} \Phi(t/2)\frac{dt}{t} + \frac{553}{100} - \frac{273}{100} + \frac{89}{50}\Phi(10).$$

Finally $\Phi(t/2) = \frac{2}{\pi}\int_0^{\pi/2} e^{-t\sin^2 u} du$. Using $\sin u \ge \frac{23}{24}u$ on $[0, \frac{1}{2}]$ and $\sin u \ge \frac{23}{48}$ on $[\frac{1}{2}, \frac{\pi}{2}]$,

$$\Phi(t/2) \le \frac{24}{23\sqrt{\pi t}} + e^{-t(23/48)^2}, \qquad \int_{40}^{\infty} \Phi(t/2)\frac{dt}{t} \le \frac{48}{23\sqrt{40\pi}} + \frac{e^{-z}}{z},$$

with $z = 40(23/48)^2$, by $\mathrm{E}_1(z) \le e^{-z}/z$. Collecting terms and dividing by $\ln 2$ gives C_*, and a positive series for $I_0(10)$ certifies $\Phi(10) < 0.127833337164$, so $C_* < 4.637$. □

8. Computations

The repository contains a script that recomputes the principal constants and the finite computations stated above and prints a pass or fail line for each. Over all symmetric functions with $n \le 16$ (exact integer Fourier coefficients, floating-point logarithms) it finds:

quantity	maximum over symmetric f, $n \le 16$
H/I	3.93027, at AND/OR with $n = 16$
L/I	2.51755
B/I	1.80986
Δ_1/I_1	4.86767, at OR_3

In every dimension $2 \le n \le 16$ the maximum of H/I is attained by AND and OR. The script also checks the integral C_N by adaptive quadrature for $N \le 200$ and for N up to 10^6, Lemma 5.2 on dense grids, Theorem B on random symmetric derivatives with $n = 20, 40, 80$, the closed form of Theorem C, and the constants C_{sym} and C_* at 40 digits. Two further scripts produce the interval certificates for Lemma 3.2 and for (6). These computations support but are not part of the proofs, except for those two certificates.

9. Questions

Question 9.1. Is the optimal constant for symmetric functions equal to 4?

Question 9.2. Find an explicit n_0 with $\Delta_1 \le 4.8723\,\mathsf{I}_1$ for every symmetric f on $n \ge n_0$ variables. Together with a computation for $n < n_0$, Lemma 7.1 would then give $\mathsf{H} \le 4.8723\,\mathsf{I}$ for all symmetric f.

Question 9.3. Can the level bound $L \le 2.692\,\mathsf{I}$ be improved for symmetric functions? Every gain there lowers C_{sym} directly. The largest value of L/I we found is 2.51755, at AND and OR with $n = 6$.

Appendix A. The scalar certificate for the restriction step

The interval certificate for (6) uses two elementary lower bounds for E_1. For $0 < z \le 1$,

$$\mathrm{E}_1(z) \ge -0.578 - \ln z + \sum_{k=1}^{8} \frac{(-1)^{k+1} z^k}{k\,k!},$$

since an even truncation of the alternating series is a lower bound and $\gamma < H_{10} - \ln 10 - \frac{1}{20} + \frac{1}{1200} < 0.578$ by the Euler–Maclaurin remainder. For $z > 1$,

$$\mathrm{E}_1(z) \ge e^{-z}\,\frac{z^2 + 8z + 11}{z^3 + 9z^2 + 18z + 6}.$$

Indeed $\mathrm{E}_1(z) = e^{-z}\,\mathbb{E}[(z+Y)^{-1}]$ for Y exponential with mean one, and for every quadratic polynomial P we have $(z+y)^{-1} \ge 2P(y) - (z+y)P(y)^2$, the difference being $[1-(z+y)P(y)]^2/(z+y)$. In the orthonormal basis 1, $1-y$, $1-2y+y^2/2$ the compression of multiplication by $z+y$ to quadratic polynomials is

$$M = \begin{pmatrix} z+1 & -1 & 0 \\ -1 & z+3 & -2 \\ 0 & -2 & z+5 \end{pmatrix},$$

and optimizing over P gives $(M^{-1})_{00}$, which is the displayed rational function. The entries of M follow from $\mathbb{E}Y^k = k!$.

Both Ψ and U decrease: h increases, and $U'(x) = -e^{-40x}/x - 5.46 - 35.6\,e^{-20x} < 0$. So an outward interval lower bound for $U(r) - \Psi(l)$ proves (6) on a whole cell $[l, r]$. For $0 < x \le 10^{-4}$ the bounds in the proof give

$$U(x) - \Psi(x) \ge 5.53 + 1.78 - 0.578 - \ln 40 - 3 - (5.46 + 20 \cdot 1.78)\,10^{-4} > 0.$$

The value $\Phi(10)$ is enclosed using $I_0(10) = \sum_{k\ge 0} 25^k/(k!)^2$, with the terms through $k = 60$ summed and the rest bounded by a geometric series of ratio at most $25/62^2$. The resulting bound is $C_* < 4.636423665924$.

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This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [9] and working together inside the Brain framework [11], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [10], the Math Brain repository [9] and the Brain framework repository [11].

Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics, and its results were developed with the Brain [11] and Math Brain [9]. In November 2026, Claude Opus 5.5 (Anthropic) and a team of Codex agents running OpenAI's GPT-6.1, all connected to the Brain, retrieved the spectral-level entropy lemma and related methods from the author's earlier work through Math Brain, sharpened the level-entropy bound with an interval certificate, derived the split of the entropy into level and within-level parts and the noise-stability estimate, proved the Bessel-function inequalities and the asymptotic restriction estimate, carried out the finite computations, and searched the literature. Separate agents refereed the proof of the main theorem and of the asymptotic restriction estimate. Claude Opus 5.5 coordinated the agents, assembled the proofs, prepared the manuscript, and audited it, including the proofs, the numerical constants and the finite computations. These checks are not peer review. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Data availability

`https://github.com/yspennstate/symmetric-fei-constant`

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [9].

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